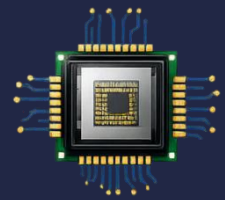


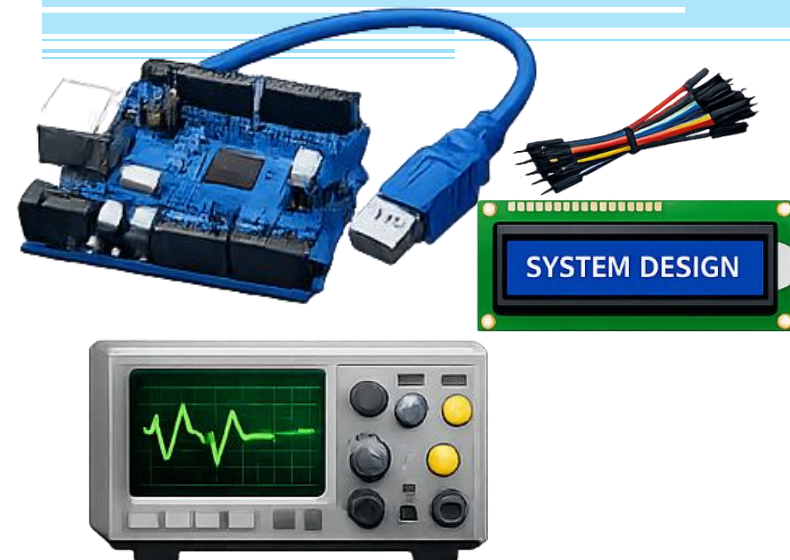
CS-301

Microprocessor- Based System Design



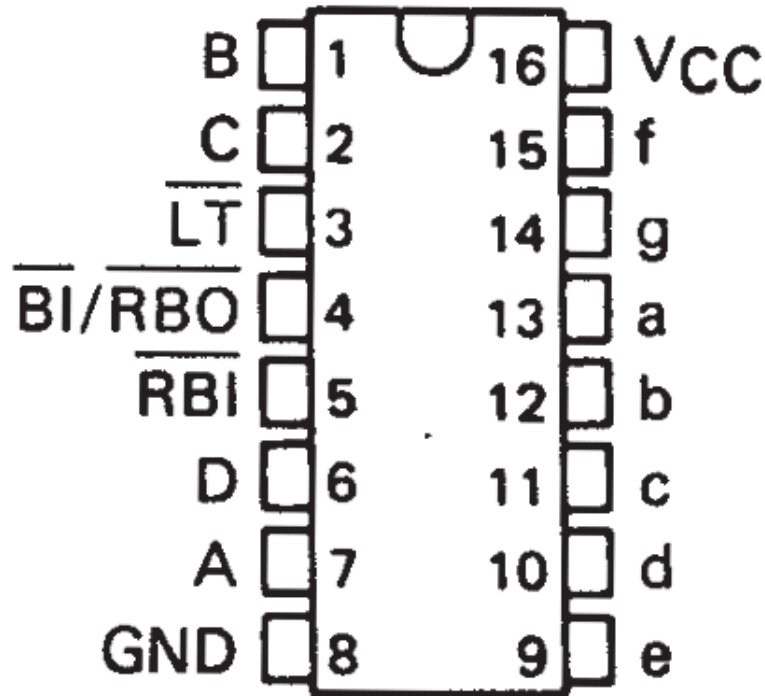
Lecture 6: Interfacing Microprocessors with Basic IO Devices(Seven-Segment Displays with BCD Decoder)

Course Teacher: Syeda Ramish Fatima

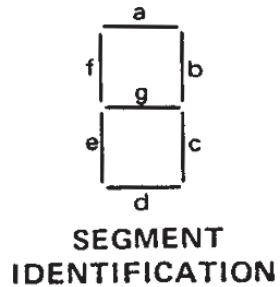


BCD to Seven Segment Decoder IC

(TOP VIEW)



BCD to Seven Segment Decoder IC



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

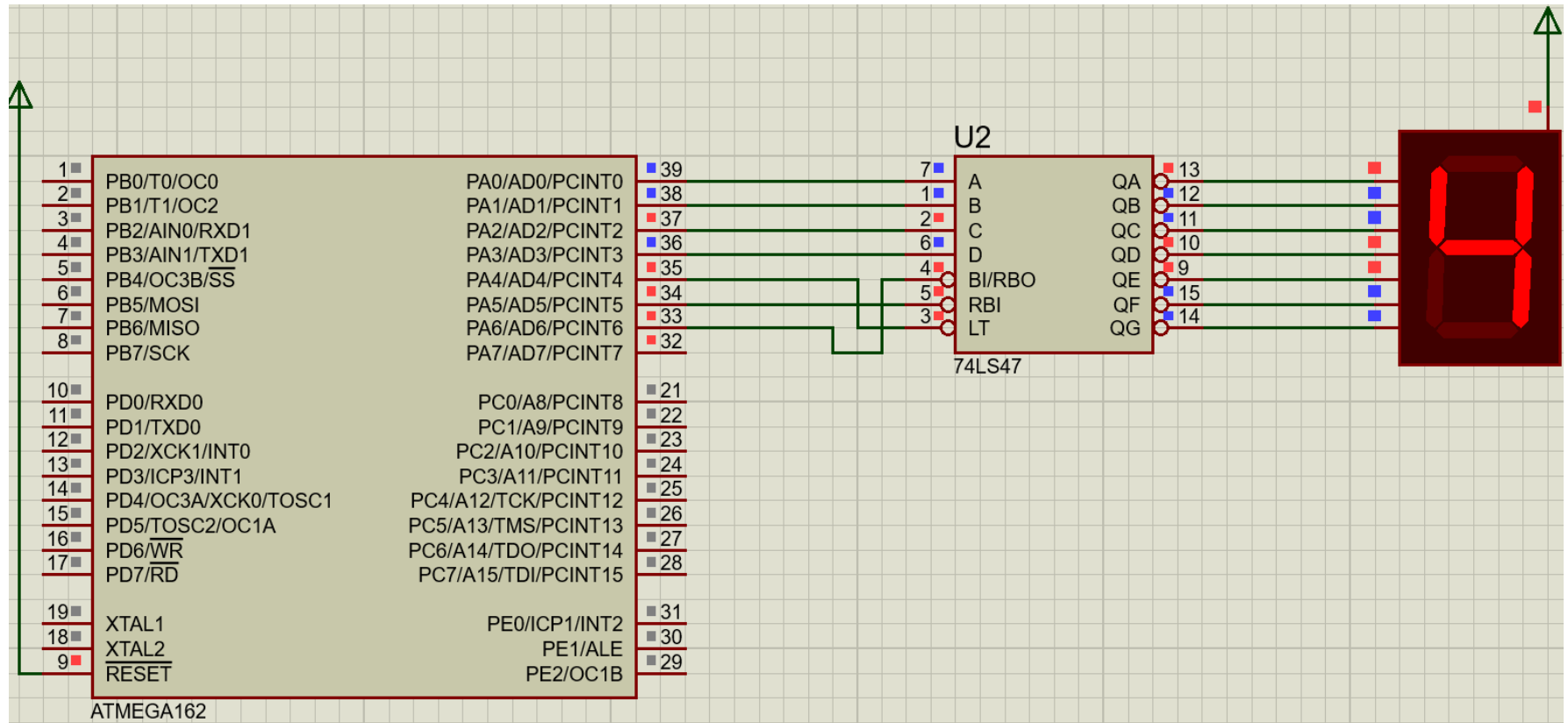
NUMERICAL DESIGNATIONS AND RESULTANT DISPLAYS

BCD to Seven Segment Decoder IC

'46A, '47A, 'LS47 FUNCTION TABLE (T1)

DECIMAL OR FUNCTION	INPUTS						$\overline{\text{BI}}/\overline{\text{RBO}}^\dagger$	OUTPUTS							NOTE
	$\overline{\text{LT}}$	$\overline{\text{RBI}}$	D	C	B	A		a	b	c	d	e	f	g	
0	H	H	L	L	L	L	H	ON	ON	ON	ON	ON	ON	OFF	1
1	H	X	L	L	L	H	H	OFF	ON	ON	OFF	OFF	OFF	OFF	
2	H	X	L	L	H	L	H	ON	ON	OFF	ON	ON	OFF	ON	
3	H	X	L	L	H	H	H	ON	ON	ON	ON	OFF	OFF	ON	
4	H	X	L	H	L	L	H	OFF	ON	ON	OFF	OFF	ON	ON	
5	H	X	L	H	L	H	H	ON	OFF	ON	ON	OFF	ON	ON	
6	H	X	L	H	H	L	H	OFF	OFF	ON	ON	ON	ON	ON	
7	H	X	L	H	H	H	H	ON	ON	ON	OFF	OFF	OFF	OFF	
8	H	X	H	L	L	L	H	ON	ON	ON	ON	ON	ON	ON	
9	H	X	H	L	L	H	H	ON	ON	ON	OFF	OFF	ON	ON	
10	H	X	H	L	H	L	H	OFF	OFF	OFF	ON	ON	OFF	ON	
11	H	X	H	L	H	H	H	OFF	OFF	ON	ON	OFF	OFF	ON	
12	H	X	H	H	L	L	H	OFF	ON	OFF	OFF	OFF	ON	ON	
13	H	X	H	H	L	H	H	ON	OFF	OFF	ON	OFF	ON	ON	
14	H	X	H	H	H	L	H	OFF	OFF	OFF	ON	ON	ON	ON	
15	H	X	H	H	H	H	H	OFF	OFF	OFF	OFF	OFF	OFF	OFF	
BI	X	X	X	X	X	X	L	OFF	OFF	OFF	OFF	OFF	OFF	OFF	2
RBI	H	L	L	L	L	L	L	OFF	OFF	OFF	OFF	OFF	OFF	OFF	3
LT	L	X	X	X	X	X	H	ON	ON	ON	ON	ON	ON	ON	4

Seven Segment Displays



Interfacing with Seven segment displays(Common Anode) with Decoder

```
#include <avr/io.h>
#define F_CPU 1000000UL
#include <util/delay.h>
int main(void)
{
    DDRA=0b11111111; // set port A as output.
    char value=4; // to assign to PORTD to drive 7-segment LEDS
    while(1)
    {
        PORTA=0b11110100; //PORTA register is being assigned the value to drive BCD
        Decoder
    } // while ends
} //main ends
```

Intefacing of ATmega162 with LCD Controller



16x2 LCD Module



16x2 LCD Module Pinout